

JANUS Photonic Amplification Stage

GaP Parametric Amplifier: Material Selection, Design Derivation, and Corrected Power Budget

Prepared in support of Patent Application No. 202611052791 — Deepanshu Bhardwaj

Document Date: July 2026

Purpose of this document

This document consolidates a full engineering trace of the JANUS optical amplification stage: why the discrete SOA baseline was rejected, why GaP was selected as the nonlinear gain medium over five competing materials, the original design flaw that made the initial '5 microwatt pump for 30 dB gain' target physically unreachable, every architecture explored to close that gap, and the final locked specification with its power budget. All calculations are re-derivable from the equations given; all literature figures are cited with their source.

Table of Contents

1. Executive Summary
2. Background: Why Amplification Is Needed in JANUS
3. The SOA Baseline and Why It Was Rejected
4. Ring Resonator Material Screening
5. The Original Design and Its Flaw
6. Root Cause: The Q-vs-Bandwidth Wall
7. Geometry Study: Radius, Buildup, and Effective Length
8. Wavelength Architecture: Single-Wavelength vs. 775/1550 nm
9. Hybrid High-Q Host Exploration
10. Pipelining: Where Amplification Sits in the Datapath
11. Final Locked Specification
12. Alternatives Considered and Rejected
13. Open Problems and Next Steps
14. References

1. Executive Summary

The JANUS photonic-electronic accelerator requires optical signal amplification at every routing/computation site to overcome the 1:64 passive splitting loss (-18.1 dB) inherent to its fan-out architecture. An early design target specified a GaP microring parametric amplifier delivering 30 dB of gain from approximately 5 microwatts of external CW pump power, at a ring quality factor (Q) of 2,000–5,000. This document records how that target was tested against first-principles nonlinear-optics physics, found to be unreachable as originally specified, and replaced with a corrected, literature-grounded design.

The core finding: at $Q=2,000$ – $5,000$, the ring's spectral bandwidth is set by the need to pass an 80–100 GHz signal pulse train without distortion. This bandwidth requirement caps achievable buildup to single-digit dB, three orders of magnitude short of what 30 dB of parametric gain requires from microwatt-level pump power. No amount of radius tuning recovers this gap, because the buildup factor and the total cavity interaction length trade off against each other in a way that keeps the achievable gain bounded.

The resolution came from re-ordering the datapath: performing amplification on the continuous-wave (CW) optical carrier before pulse formation (via the TFLN Mach-Zehnder modulator and chirped Bragg grating compressor), rather than after. This removes the bandwidth constraint on the amplifying ring entirely, since a CW carrier has no data-rate requirement to preserve. It also enables consolidating amplification before the 1:64 fan-out rather than after — reducing the number of amplifier instances from roughly 600,000 to 8,192, a proportional reduction in total system pump power.

The final locked specification is a pure GaP microring, single-wavelength operation near 1550 nm (avoiding two-photon absorption, which reactivates below $\sim 1,097$ nm in GaP), radius 5 micrometers, and $Q \approx 290,000$ – $300,000$ — a figure anchored to a real demonstrated GaP-on-insulator ring resonator, not an assumed or aspirational value. At this operating point, 21 dB of gain (sufficient to cover the 1:64 split loss with margin) requires approximately 400 microwatts of pump power per site, and the full amplification stage across 8,192 sites totals approximately 3.3–3.5 W — roughly 3% of the JANUS datacenter configuration's total system power budget.

2. Background: Why Amplification Is Needed in JANUS

JANUS routes optical signals through a 15-stage PCM (phase-change material) Beneš fabric to implement reconfigurable spatial matrix computation. Before entering this fabric, each signal is passively split 1:64 to fan out across the computation array. A passive 1:64 splitter introduces an unavoidable insertion loss of:

$$Loss(dB) = 10 \cdot \log_{10}(1/64) \approx -18.1 \text{ dB}$$

This loss must be recovered somewhere in the optical chain before the signal reaches the Ge-fin photodetector, or the detection margin collapses. The JANUS architecture's original specification called for a per-site parametric amplifier delivering 30 dB of gain — enough to cover the split loss with roughly 12 dB of margin for routing and coupling losses elsewhere in the fabric.

3. The SOA Baseline and Why It Was Rejected

The first candidate evaluated for this amplification role was a conventional discrete Semiconductor Optical Amplifier (SOA), the incumbent technology in telecom optical links.

SOA class	Electrical power draw	Notes
High-power discrete (BOA1554P-class)	~1.8 W per device	I≈1200 mA, V≈1.5 V; 4-SOA module draws 7.2 W
Low-power variant (10–15 dB gain)	~50 mW per device	Lower bias current, reduced max gain
AWGR photonic-accelerator integrated SOA	42–84 mW per device	Scales with data rate and gain configuration

At JANUS's scale of roughly 600,000 amplification sites (pre-consolidation), even the lowest-power SOA class evaluated (50 mW) would require:

$$600,000 \times 50 \text{ mW} = 30,000 \text{ W (30 kW)}$$

This is categorically incompatible with a chip-scale accelerator power budget. SOAs also introduce electrical-to-optical conversion inefficiency, amplified spontaneous emission (ASE) noise, and gain-recovery times on the order of hundreds of picoseconds (carrier recombination-limited) — all of which motivated moving to an all-optical, cavity-enhanced parametric gain mechanism instead, where gain is instantaneous (femtosecond-scale, $\chi^{(2)}/\chi^{(3)}$ electronic-cloud polarization response) and driven by optical rather than electrical pump power.

4. Ring Resonator Material Screening

The nonlinear gain medium was selected against three simultaneous criteria: (1) Pockels/EO performance at or above thin-film lithium niobate (TFLN), (2) strong Kerr ($\chi^{(3)}$) nonlinearity for

parametric gain, and (3) two-photon-absorption-free (TPA-free) operation at the signal wavelength. Six candidate materials were evaluated.

Material	Verdict
TFLN (LiNbO ₃)	Strong EO (Pockels) performance; used elsewhere in JANUS (MZM, ring routers) but weaker intrinsic Kerr gain per unit pump than GaP
AlN	Excellent CTE match to GaP and native $\chi^{(2)}$; Pockels coefficient weak ($\sim 1\text{--}1.5$ pm/V pure, $\sim 3\text{--}10$ pm/V Sc-doped) — see Section 9 for its role as a high-Q host
BaTiO ₃	Recommended in a separate (non-JANUS) photonics screening for its coupled double-ring bistability profile; not carried into the JANUS PSA decision
GaP	Selected — see below
Si	TPA-active at telecom wavelengths under high intensity; ruled out early
AlGaAs	Strong nonlinearity but narrower TPA-free window than GaP; not pursued further

4.1 Why GaP

- Kerr index $n_2 \approx 1.2 \times 10^{-17} \text{ m}^2/\text{W}$ — among the highest of CMOS/foundry-compatible integrated platforms, roughly 3× stronger than the GeSbS chalcogenide alternative evaluated in Section 9.
- TPA-free for photon energies below its indirect bandgap (2.26 eV) when combined two-photon energy stays under that threshold — satisfied at 1550 nm ($2 \times 0.80 \text{ eV} = 1.60 \text{ eV}$, safely below gap) but not at 775–800 nm ($2 \times 1.55\text{--}1.60 \text{ eV} = 3.10\text{--}3.20 \text{ eV}$, well above gap). This becomes decisive in Section 8.
- Real, demonstrated integration on silicon-compatible substrates (GaP-on-insulator), with published ring resonator Q factors and second-harmonic-generation conversion efficiency data available as calibration anchors (Logan et al.; Wilson et al. — see References).
- Second-order ($\chi^{(2)}$) nonlinearity also present ($d_{\text{eff}} \approx 70 \text{ pm/V}$), enabling an alternative pump-degenerate architecture explored and ultimately rejected in Section 8.

5. The Original Design and Its Flaw

The original JANUS specification called for a GaP microring with $Q > 10,000$ for phase-sensitive amplification, later corrected (in the project's own internal error analysis) to $Q \approx 2,000$ with 5 ps pulses, to spectrally match a 100 GHz pulse train. Independently, a design target of ~ 5 microwatts of external CW pump power producing 30 dB of parametric gain was carried forward at this Q. This section documents how that target was tested and where it failed.

5.1 The bandwidth-matching constraint that set $Q \approx 2,000\text{--}2,750$

A transform-limited Gaussian pulse of duration τ has spectral bandwidth $\Delta f_{\text{pulse}} \approx 0.44/\tau$. For the ring cavity to pass such a pulse without distorting or stretching it, its own passband must be at least this wide:

$$\Delta f_{\text{cavity}} = f_0/Q \geq \Delta f_{\text{pulse}} = 0.44/\tau$$

Solving for the maximum Q at several candidate pulse/clock rates (assuming pulse width $\approx$ half the clock period):

Pulse/clock rate	Q_max (bandwidth-limited)
20 GHz	$\sim 11,000$
33 GHz	$\sim 6,700$
50 GHz	$\sim 4,400$
80 GHz	$\sim 2,750$
100 GHz	$\sim 2,200$

This confirmed the project's own $Q \approx 2,000$ correction was internally consistent for 80–100 GHz operation — but it also set a hard ceiling that could not be raised without dropping the pulse rate.

5.2 Buildup at this Q is far too small for 30 dB

Resonant power buildup in a critically-coupled ring is set by the cavity finesse:

$$\text{Finesse } F = Q/m, \quad m = n_g \cdot L / \lambda_o \quad (\text{mode number}), \quad \text{Buildup } B \approx F/\pi$$

For a 20 micrometer radius GaP ring ($n_g \approx 3.3$, $\lambda_o = 1550$ nm), $m \approx 267$, giving at most ~ 4 – 11 dB of buildup across the Q range the bandwidth constraint allows (Section 6 develops this fully). This is nowhere close to the ~ 30 dB the design target required from resonant enhancement alone.

5.3 Radius does not close the gap — it was tested exhaustively

A natural hypothesis was that shrinking the ring radius could recover the missing gain, since smaller rings have fewer modes (m) and therefore higher finesse for fixed Q. This was tested down to $R = 1$ – 2 micrometers. The result: total effective interaction length inside the cavity,

$$L_{\text{eff}} = N_{\text{roundtrips}} \times L = Q \cdot \lambda_o / (2\pi \cdot n_g)$$

depends only on Q, not on radius — the radius-dependence in the number of round trips exactly cancels the radius-dependence in the round-trip length. Smaller radius does increase peak circulating buildup (roughly 4 dB gained from $5 \mu\text{m} \rightarrow 2 \mu\text{m}$), but total achievable parametric gain (which depends on both circulating intensity and total interaction length) still falls far short of 30 dB at the bandwidth-limited Q, and sub- $3 \mu\text{m}$ radii approach fabrication limits where bend-radiation loss would likely erode Q before any gain benefit is realized.

Conclusion of Section 5

At the Q the 80–100 GHz pulse rate forces ($\sim 2,000$ – $2,750$), no realistic combination of radius, coupling length, or pump power within a single ring reaches 30 dB of parametric gain from microwatt-level pump. First-principles calculation using GaP's known Kerr index put the actual pump requirement at approximately 5–10 W for 15–30 dB at this Q — roughly six orders of magnitude above the 5 microwatt target.

6. Root Cause: The Q-vs-Bandwidth Wall

The underlying problem is that a single ring resonator is being asked to do two incompatible jobs simultaneously: pass a wideband pulsed data signal (requiring low Q) and build up enough circulating intensity for strong parametric gain (requiring high Q). These pull in opposite directions on the same parameter.

6.1 Quantitative pump power at the bandwidth-limited Q

Using the standard undepleted-pump parametric gain formula for four-wave mixing,

$$G(\text{dB}) = 10 \cdot \log_{10}(\cosh^2(\gamma \cdot P_{\text{circ}} \cdot L_{\text{eff}})), \quad \gamma = 2\pi \cdot n_2 / (\lambda \cdot A_{\text{eff}})$$

with GaP's $n_2 \approx 1.2 \times 10^{-17} \text{ m}^2/\text{W}$ and a representative mode area $A_{\text{eff}} \approx 0.2 \text{ } \mu\text{m}^2$, the pump power required at $R=5 \text{ } \mu\text{m}$, $Q \approx 2,200$ was calculated directly:

Target gain	Required external pump (single-ring, bandwidth-limited Q)
10 dB	~4.3 W
15 dB	~5.8 W
30 dB	~9.9 W

This result was cross-checked using an alternative (and looser) effective-length approximation sometimes used informally ($L_{\text{eff}} \approx Q \cdot 2\pi R$, treating Q as if it equaled the number of physical round trips directly). Even this generous approximation — which overstates the true effective length by a factor of $2\pi \cdot m$ ($\approx 420\times$ at $R=5 \text{ } \mu\text{m}$) — still requires pump power in the low tens of milliwatts, not microwatts, confirming the gap was not merely a modeling artifact.

6.2 The resolution: decouple the signal path from the pump-buildup path

The wall is only present because the ring was assumed to carry the already-pulsed 80–100 GHz signal. If amplification instead acts on the CW optical carrier before pulse formation — i.e., before the TFLN Mach-Zehnder modulator and chirped Bragg grating compressor stage that creates the 5 ps pulses — the ring never needs to pass wideband pulsed content. A CW tone has no bandwidth requirement beyond sitting on resonance, so Q is free to rise to whatever value the fabrication process actually supports, with no bandwidth penalty.

This single re-ordering — amplify-then-pulse rather than pulse-then-amplify — is what makes the rest of this document's final specification possible.

7. Geometry Study: Radius, Buildup, and Effective Length

With the bandwidth constraint removed (Section 6.2), Q is no longer capped at ~2,750. The remaining design question is what combination of radius and Q is both physically realistic and pump-efficient. This section summarizes the systematic sweep performed.

7.1 Buildup vs. radius, fixed Q

For a fixed Q, buildup scales as $1/m$, i.e., inversely with radius (fewer modes in a smaller ring raises finesse). This was confirmed numerically:

Radius	Mode number (m), Q=2,000	Buildup (dB)
20 μm	268	7.7
10 μm	134	10.8
5 μm	67	13.8
2 μm	27	17.9 (fabrication risk — see below)

However, because total effective interaction length is radius-independent (Section 5.3), this buildup gain alone was insufficient to reach 30 dB at bandwidth-limited Q. It becomes relevant again once Q is freed from the bandwidth constraint (Section 7.2).

7.2 Fabrication floor

Radii below approximately 3–5 μm risk bend-radiation loss dominating over material/scattering loss, which would degrade Q below the value assumed in any buildup calculation. R=5 μm was retained as the practical floor because it coincides with the radius of the real, demonstrated GaP-on-insulator ring resonator used as the Q calibration anchor for the final specification (Section 11).

7.3 Final geometry sweep, single-wavelength, real/near-real Q

Once the amplifier was moved to the CW pre-pulse position (removing the bandwidth cap) and the wavelength question was resolved to single-wavelength ~1550 nm operation (Section 8), the following sweep was performed using the $\chi^{(3)}$ Kerr gain formula from Section 6.1:

Radius	Q	Pump for 21 dB / site	Total, 8,192 sites
10 μm	290,000 (measured)	854 μW	7.0 W
10 μm	300,000 (near-term)	798 μW	6.5 W
5 μm	290,000 (measured)	427 μW	3.5 W
5 μm	300,000 (near-term)	399 μW	3.3 W

Halving the radius from 10 μm to 5 μm at fixed Q roughly halves the required pump power, consistent with the $1/m$ buildup scaling in Section 7.1 — with total interaction length held constant by Q, the

buildup advantage translates directly into a pump-power advantage rather than being cancelled out. $R=5\text{ }\mu\text{m}$, $Q\approx 290,000\text{--}300,000$ was carried forward as the final geometry (Section 11).

8. Wavelength Architecture: Single-Wavelength vs. 775/1550 nm

Two nonlinear mechanisms were evaluated for the GaP gain stage, corresponding to two different pump-wavelength architectures.

8.1 Option 1 — $\chi^{(2)}$ degenerate optical parametric amplification (775 nm pump / 1550 nm signal)

GaP's second-order nonlinearity ($d_{\text{eff}}\approx 70\text{ pm/V}$) supports parametric down-conversion from a 775 nm pump to a degenerate 1550 nm signal/idler pair. This mechanism was attractive because 775–800 nm diode lasers can reach wall-plug efficiencies above 60%, substantially higher than typical 1550 nm telecom laser WPE ($\sim 30\text{--}40\%$).

However, checking this against GaP's bandgap structure reveals a disqualifying problem: two-photon absorption.

$$2 \times E_{\text{photon}}(800\text{ nm}) = 3.10\text{ eV} > E_{\text{gap,indirect}}(\text{GaP}) = 2.26\text{ eV}$$

At 775–800 nm, the combined two-photon energy exceeds both GaP's indirect (2.26 eV) and direct (2.78 eV) bandgaps — TPA is strongly active. GaP only remains TPA-free for wavelengths longer than approximately 1,097 nm. Operating the pump at 775–800 nm reintroduces the exact loss mechanism that was one of the three original screening criteria for selecting GaP over competing materials (Section 4).

This mechanism was quantified anyway, for comparison. Using the coupled-wave OPA gain formula (Boyd normalization) with the stated d_{eff} , mode area, and coupling geometry, a pure GaP ring at its real demonstrated Q ceiling (290,000) required approximately 30.8 mW per site for 21 dB gain — roughly 36× more pump power than the equivalent single-wavelength $\chi^{(3)}$ design at the same Q and radius, precisely because χ^2 gain at this Q only accumulates over ~ 345 recirculation passes versus the much longer effective interaction length available to the χ^3 path. The 775 nm/ $\chi^{(2)}$ path was rejected on these combined grounds — TPA risk and higher realized pump power — despite its laser-efficiency advantage.

8.2 Option 2 — $\chi^{(3)}$ Kerr four-wave mixing, single-wavelength (~ 1550 nm pump and signal)

This is the mechanism carried into the final design. Pump and signal both sit near 1550 nm (non-degenerate FWM, small detuning between pump/signal/idler), keeping every photon below GaP's TPA threshold and preserving the JANUS architecture's core single-wavelength invariant (which independently exists to suppress FWM/XPM cross-talk in the routing fabric — see JANUS_Error_Analysis_and_Corrections, Section 8.2). This is the mechanism used for all final power-budget figures in Section 11.

8.3 Ge-fin photodetector compatibility check

Because the 775/800 nm option was under consideration, the downstream Ge-fin photodetector (265 GHz bandwidth, Lischke et al. 2021 — the detector already specified elsewhere in JANUS) was checked for compatibility at that wavelength. Germanium's absorption coefficient is strong from the visible through 1550 nm and only begins dropping sharply above ~1550 nm — meaning 775–800 nm sits well within Ge's high-absorption band, and a thinner absorption layer (feasible at this wavelength) would if anything favor higher bandwidth via reduced carrier transit time. No literature demonstration of the specific 265 GHz design re-optimized for 800 nm was found, but no physical obstacle was identified either. This check became moot once the 775 nm pump path was rejected on TPA grounds (Section 8.1), but is retained here as it may be relevant if a $\chi^{(2)}$ path is revisited with a different, TPA-safe host material (see Section 13).

9. Hybrid High-Q Host Exploration

Before settling on the pure-GaP, real-Q-anchored design (Section 11), a parallel line of investigation explored whether a separate ultra-high-Q host material (evanescently coupled to a thin GaP nonlinear layer) could beat pure GaP's own Q ceiling. This section is retained because it surfaced a materially important methodological finding, and because it may become the better path if GaP's own fabrication ceiling does not improve (see Section 13).

9.1 Candidate host materials

Material	Best demonstrated Q (2025–2026 lit.)	Notes
Si ₃ N ₄	~1.1×10 ⁷ (subtractive, foundry); >2×10 ⁸ (absorption-limited, ultralow-loss wafer-scale)	Most mature ultra-high-Q integrated platform
TFLN / LiNbO ₃	1.4×10 ⁷ (vertical evanescent coupling to SiN, 2025); 4×10 ⁷ (all-LN microdisk, PLACE)	Already present elsewhere in JANUS material stack
AlN (single-crystal, sapphire)	2.8×10 ⁶ (2025)	CTE-matched to GaP; reintroduces sapphire CTE mismatch
AlN (racetrack, optimized)	5.4×10 ⁶ (Jan 2026)	Foundry-compatible, no sapphire substrate
GeSbS (chalcogenide)	2–5×10 ⁶ (R≈100 μm)	Real demonstrated 31.5 dB gain at 8.5 mW pump — but this is the material itself acting as the nonlinear medium, not a GaP host (see 9.3)

9.2 The citation-mismatch finding

An early version of the internal design reasoning cited "Xiao et al., OFC 2024, Th1D.4 — GaP integrated OPA, 31.5 dB gain at 8.5 mW pump" as a calibration anchor. Checking this citation directly found it does not describe a GaP device at all — Th1D.4 (Xiao, Xia, Luo, Li, Zhang) demonstrates a GeSbS microresonator. The real demonstrated GaP parametric amplifier in the 2024–2025 literature (Kuznetsov et al.) is an unrelated device type: a 5.55 cm traveling-wave spiral waveguide with no resonant cavity, reaching 35 dB gain via long propagation length and dispersion engineering rather than resonant buildup, at substantially higher (~100+ mW) pump power. Neither cited paper is a valid GaP-microring calibration point, and this was corrected before further calculations relied on it.

9.3 GeSbS is a real high-Q, low-power result — but at the wrong Q for JANUS's original bandwidth constraint

The real GeSbS result ($Q \approx 2 \times 10^6$ – 5×10^6 , $R \approx 100 \mu\text{m}$, 31.5 dB at 8.5 mW) was checked against the 80 GHz bandwidth requirement that originally constrained the signal-path ring (Section 5.1). Its cavity bandwidth (39–97 MHz) is roughly 1,000× narrower than the ~70–90 GHz an 80 GHz pulse train requires. This confirmed that no single-ring material choice resolves the bandwidth-vs-gain conflict — the constraint is architectural (Section 6.2), not a materials problem, which was the basis for moving amplification to the CW pre-pulse position.

9.4 Hybrid GaP-on-host evanescent coupling: quantified but unverified

With amplification moved to the CW carrier, a GaP-nonlinear-layer-on-high-Q-host architecture was quantified (TFLN host, $Q = 1.4 \times 10^6$, GaP modal overlap 30–44% based on a comparable Si/III-V hybrid confinement demonstration): approximately 42–72 μW pump per site for 30 dB, or roughly 50 μW at a 44% overlap figure. This 44% figure, however, was sourced from an unrelated device type (a Si/III-V gain waveguide, not a resonator, and not a GaP/TFLN pairing) — no published device has demonstrated high-Q and high-GaP-overlap simultaneously in one structure. This combination is flagged as an aspirational, not validated, design point (Section 13).

A further sweep tested whether pushing this hybrid host to $Q \approx 10^7$ (real Si_3N_4 /TFLN ceiling) could reach single-digit microwatt pump power. It does (≈ 1 – $2 \mu\text{W}$ at $Q = 10^7$, 30% overlap) — but the hybrid integration, bonding-interface loss, and dual-material CTE/lattice matching this requires were judged higher-risk than the pure-GaP path that Section 11 ultimately adopted.

10. Pipelining: Where Amplification Sits in the Datapath

10.1 Original position: post-fanout, one amplifier per output

The original architecture placed one amplifier at each of the ~600,000 post-1:64-split sites, after pulse formation. This is the configuration that ran into the Q-vs-bandwidth wall (Section 6) and, even after that wall was resolved, would still require 600,000 individually pump-delivered and individually EO-locked amplifier instances.

10.2 Corrected position: pre-fanout, on the CW carrier, before pulse formation

Two changes were made together:

- Amplification acts on the CW laser carrier, upstream of the TFLN MZM + chirped Bragg grating pulse-compression stage (which forms the 5 ps / 80–100 GHz pulse train) and upstream of the EAM data-modulation stage. This removes the bandwidth constraint (Section 6.2).
- Amplification is consolidated before the 1:64 fan-out, at the laser-count level (8,192 sites, matching the JANUS datacenter configuration's laser count) rather than after it (600,000 sites). Since the 21 dB gain target exists to cover the 1:64 split loss regardless of where in the chain it is applied, and since per-amplifier pump power does not depend on how many downstream channels that one amplified signal eventually feeds, moving amplification upstream of the split reduces the amplifier count by the split ratio (64×, partially offset by the 8,192-vs-600,000 accounting) without any loss of coverage.

This reordering was checked for two potential problems, both resolved favorably:

- Timing: because both pump and signal are CW at the point of amplification, the ring's photon lifetime ($\tau_{ph} = Q_L / \omega$) manifests only as a one-time cavity-fill startup delay, not a per-symbol throughput bottleneck. All picosecond/GHz-scale data timing (EAM, MZM) occurs downstream of the amplifier and is unaffected by the ring's cavity dynamics.
- Optical intensity handling: consolidating amplification pre-fanout means each amplifier's output must supply 64× more downstream channels, raising absolute circulating/output optical power per site. This raises a legitimate follow-up question — whether this pushes the design toward pump depletion or (given GaP's own TPA edge) an intensity regime requiring re-examination — flagged as an open item in Section 13.

11. Final Locked Specification

Parameter	Value
Nonlinear gain medium	Pure GaP microring (no hybrid host material)
Nonlinear mechanism	$\chi^{(3)}$ Kerr four-wave mixing, single-wavelength (~1550 nm pump and signal)
Ring radius	5 μm
Loaded Q	$\approx 290,000$ (real, demonstrated GaP-on-insulator ring) to 300,000 (near-term projection)
Target gain per amplifier	21.0 dB (exactly covers 1:64 split loss of -18.1 dB, ~ 2.9 dB margin)
Amplifier position in datapath	Pre-pulse-formation (on CW carrier), pre-1:64-fanout
Number of amplifier instances	8,192 (one per laser source, matching datacenter laser count)
Pump power per site (Q=290,000)	≈ 427 μW
Pump power per site (Q=300,000)	≈ 399 μW
Total amplification-stage power (Q=290,000)	≈ 3.5 W
Total amplification-stage power (Q=300,000)	≈ 3.3 W
Share of total JANUS datacenter system power (~107–127 W)	$\approx 3\%$

Parameter	Value
TPA status at operating wavelength	TPA-free ($2 \times 0.80 \text{ eV} = 1.60 \text{ eV}$, below GaP's 2.26 eV indirect gap)

Why this is the strongest-grounded result in the design trace

Every other configuration considered in this document (hybrid AlN/SiN host, χ^2 775 nm pump, unverified $Q=3 \times 10^6$ pure GaP) relied on at least one unverified assumption — either an un-demonstrated Q , a mismatched literature citation, or a materials combination with no published device. The final specification uses only a real, measured GaP ring Q , a single wavelength consistent with GaP's own TPA-free window, and an architecture (pre-pulse CW amplification) whose timing behavior was independently checked rather than assumed.

12. Alternatives Considered and Rejected

Alternative	Why rejected
Discrete SOA array	30 kW at 600k sites even at the lowest-power SOA class evaluated; ASE noise; hundreds-of-ps gain recovery time
Single ring, $Q=2,000$ – $2,750$, post-fanout, post-pulse (original spec)	Bandwidth-limited buildup insufficient by ~ 6 orders of magnitude in pump power for 30 dB gain (Section 5–6)
Sub-5 μm radius (2–3 μm) to boost buildup further	Diminishing returns ($\sim 4 \text{ dB}$ gain from $5 \rightarrow 2 \mu\text{m}$) plus fabrication risk: bend-radiation loss likely erodes Q below assumed values at these radii
Cascaded 2–3 ring stages at moderate Q , post-fanout	Noise-cascade analysis (Friis formula) showed negligible noise penalty for cascading, but $\pm 2 \text{ nm}$ fabrication/etch tolerance versus sub-picometer cavity linewidths at high Q made independent dual-resonance phase-locking per ring a severe control-system cost (thousands of independently-locked resonance-trim loops at 600k scale)
$\chi^{(2)}$ 775 nm pump / 1550 nm signal, pure GaP	TPA strongly active at 775–800 nm in GaP (Section 8.1); also $\sim 36\times$ higher realized pump power than the χ^3 single-wavelength path at equal Q and radius
$\chi^{(2)}$ 775 nm pump, hybrid AlN host ($Q=3.78 \times 10^6$)	TPA-safe (gain occurs in AlN region assumptions) but requires a second wavelength/laser type, additional lattice/CTE-matched hybrid integration, and still costs $\sim 60 \text{ W}$ system-wide versus ~ 3.3 – 3.5 W for the adopted design
GeSbS as the nonlinear medium directly (real $Q=2$ – 5×10^6 demonstration)	Real, low-power result (31.5 dB at 8.5 mW) but only at cavity bandwidths (39–97 MHz) incompatible with any architecture still requiring the ring to pass pulsed data; also $\sim 3\times$ weaker Kerr index than GaP
Hybrid TFLN/SiN host + GaP evanescent layer, $Q=10^6$ – 10^7	Physically promising (as low as ~ 1 – $2 \mu\text{W}$ at $Q=10^7$) but requires an unverified combination of high host Q and high GaP modal overlap that has not been jointly demonstrated in one device; retained as a future path, not adopted (Section 13)

Alternative	Why rejected
Post-fanout amplifier placement (600k sites) with the corrected CW/pre-pulse timing fix	Timing-safe but power-inefficient: 64× more amplifier instances than the pre-fanout placement for no coverage benefit

13. Open Problems and Next Steps

- Optical intensity / pump depletion check at the consolidated 8,192-site, 64×-downstream-channel amplifier output — not yet verified against GaP's TPA edge or gain-saturation behavior at the higher absolute power this consolidation implies.
- Non-degenerate FWM detuning: the exact pump-signal wavelength offset needed for practical signal/idler separation, balanced against the ring's phase-matching bandwidth at $Q \approx 290,000$ – $300,000$, has not yet been derived.
- Hybrid TFLN/SiN-host + GaP-evanescent-layer path (Section 9.4) remains open as a lower-power alternative if a real device demonstrating both high host Q and high GaP modal overlap is published, or if GaP's own fabrication ceiling does not improve beyond $\sim 300,000$ in the near term.
- Bend-loss characterization specific to GaP-on-insulator at $R=5\text{ }\mu\text{m}$ has not been independently confirmed beyond the fact that this radius matches the real $Q=290,000$ demonstration's own geometry; a dedicated bend-loss-vs-radius dataset for GaP would tighten this.
- Pump routing and delivery to 8,192 sites (fiber array, on-chip splitting network, or shared laser bank) has not yet been designed — this is now a substantially smaller problem than the original 600,000-site version but still needs a concrete routing architecture.
- EO/thermal locking strategy for the final pure-GaP ring (post-fab trimming vs. active locking) has not been re-derived for this specific geometry; the AlN Pockels-based locking scheme explored in the hybrid-host thread does not directly transfer to pure GaP without re-evaluating GaP's own EO coefficient and electrode geometry at $R=5\text{ }\mu\text{m}$.

14. References

- Wilson, D. J. et al. (2020). "Integrated gallium phosphide nonlinear photonics." *Nature Photonics*, 14, 57–62.
- Logan, D. F. et al. GaP-on-oxide ring resonators: $Q \sim 10^4$ with $\sim 400\%/W$ second-harmonic-generation conversion efficiency (Princeton/UW).
- Kuznetsov, A. et al. (2024/2025). Integrated gallium phosphide travelling-wave optical parametric amplifier: 5.55 cm spiral waveguide, up to 35 dB gain. (Also presented as OFC 2024, Th1D.3.)
- Xiao, [author list unconfirmed] et al. OFC 2024, paper Th1D.4. GeSbS microresonator phase-sensitive parametric amplification, 31.5 dB gain at 8.5 mW pump. Note: this paper describes a GeSbS device, not GaP — see Section 9.2 for the citation-correction context.
- Xia, F. et al. (2022). Ge₂₅Sb₁₅Se₆₀ chalcogenide nonlinear index (n_2) characterization via Z-scan; GeSbS microring resonators, $R \approx 100\text{ }\mu\text{m}$, intrinsic Q up to $\sim 2.3 \times 10^6$ – 5×10^6 .
- Lischke, S. et al. (2021). "Ultra-fast germanium photodiode with 3-dB bandwidth of 265 GHz." *Nature Photonics*, 15, 925–931.
- Wang, C. et al. (2018). "Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages." *Nature*, 562, 101–104.

He, M. et al. (2019). "High-performance hybrid silicon and lithium niobate Mach-Zehnder modulators." *Nature Photonics*, 13, 359–364.

Hwang et al. (2025). TFLN photonic crystal Fabry-Pérot resonator via PLACE fabrication, loaded $Q = 1.4 \times 10^6$.

Peng et al. (2025). *Optics Letters*. Single-crystal AlN-on-sapphire ring resonator, record intrinsic $Q = 2.8 \times 10^6$ at 1550 nm.

Ding et al. (Jan 2026). arXiv preprint. Polycrystalline/optimized AlN racetrack resonators, $Q = 5.4 \times 10^6$ (near-IR), dispersion-engineered for AlN Dirac solitons.

Steckler, J. et al. (2024). "Ge electro-absorption modulator bandwidth characterization at 100 GHz." *IEEE Transactions on Electron Devices*, 71, 3417–3423.

JANUS Architecture — Physics-Level Error Analysis, Corrections & Verified Solutions. Internal document, Patent Application No. 202611052791, June 2026.

A note on reference verification

Several figures in this document (Section 9.2 in particular) originated from a citation that was later found to be mismatched to the wrong material system. Every reference above has been checked at least once against a live web search during this design process; however, full bibliographic detail (exact author lists, volume/page numbers) was not independently confirmed for every entry, particularly the 2025–2026 preprints. Before this document is used in an external-facing submission, each reference should be re-verified against the original source.